

Active Reading

Although I'm grouping active reading under the category of speed reading, it could be described as the opposite of speed reading. Active reading slows down your reading rate while greatly increasing your comprehension ability. Active reading goes beyond simply highlighting your textbook and writing small notes in the margins, but fully integrating ideas as they hit you.

To start active reading, begin with your book and a notepad. On the notepad write the heading for the chapter and any subheadings for the section you are reading. After finishing reading a subsection (using your pointer!), turn to your notepad and make a few notes.

During your active reading you should note down:

- 1) What the major points are from the section.
- 2) How I can remember the major points from the section.
- 3) How I can extend or apply the major points from this section.

The first question simply gets you to acquire the information fully. The second

question forces you to link, visualize or metaphor the information. The third question gets you to move beyond this and apply the information in a different context. These three questions force you to move each major point through the understanding, explore and finally application phase of holistic learning.

For example, let's say I just finished reading a chapter in psychology about classical conditioning. The major points might be:

- Discovered by Ivan Pavlov
- Connects a stimulus with a response

I might remember this by:

- Picturing Pavlov's dogs which drooled at the sound of a bell.

I might extend this by:

- Remembering how I feel compelled to answer the phone if I hear a similar ringtone on a television program.

Clearly if you did this entire active reading process on every idea in your textbook, you might become overwhelmed. Once you become familiar with it, I strive to use active reading only on sections I have trouble understanding or remembering.

Mastering Speed Reading

Speed reading, like holistic learning, is a skill. That means reading this book has no benefit to you unless you practice. Throughout this book there will be dozens of opportunities to practice and build new skills. My suggestion is to pick only one or two of these Mind Challenges at a time and practice them for a few weeks before taking on another.



Quick Tip!

Don't forget to make use of the Speed Reading bonus material. The printout can make it easier to track your practice reading trials.



Mind Challenge!

The goal of this Mind Challenge is to make you comfortable with speed reading techniques and increase your reading speed and comprehension. After you've completed this mind challenge, pick another in the book and attempt it.

- 1) Buy one or two books that are similar to the type of material you want to be able to speed read. These will be your books for practice reading.
- 2) Commit for at least 3 weeks to practice either speed or comprehension exercises (as outlined in Practice Reading) for fifteen minutes a day on your practice books.
- 3) Also commit to using your finger as a pointer for the next two weeks on all written material on paper.
- 4) Practice active reading with at least one of your textbooks once a week.

Flow-Based Notetaking



Flow-Based Notetaking

I'm not a fan of taking detailed and intricate notes. I'm a believer in the "learn it once" principle, which means you should be listening and processing the information as your instructor is teaching—not just transcribing it on a piece of paper to learn later.

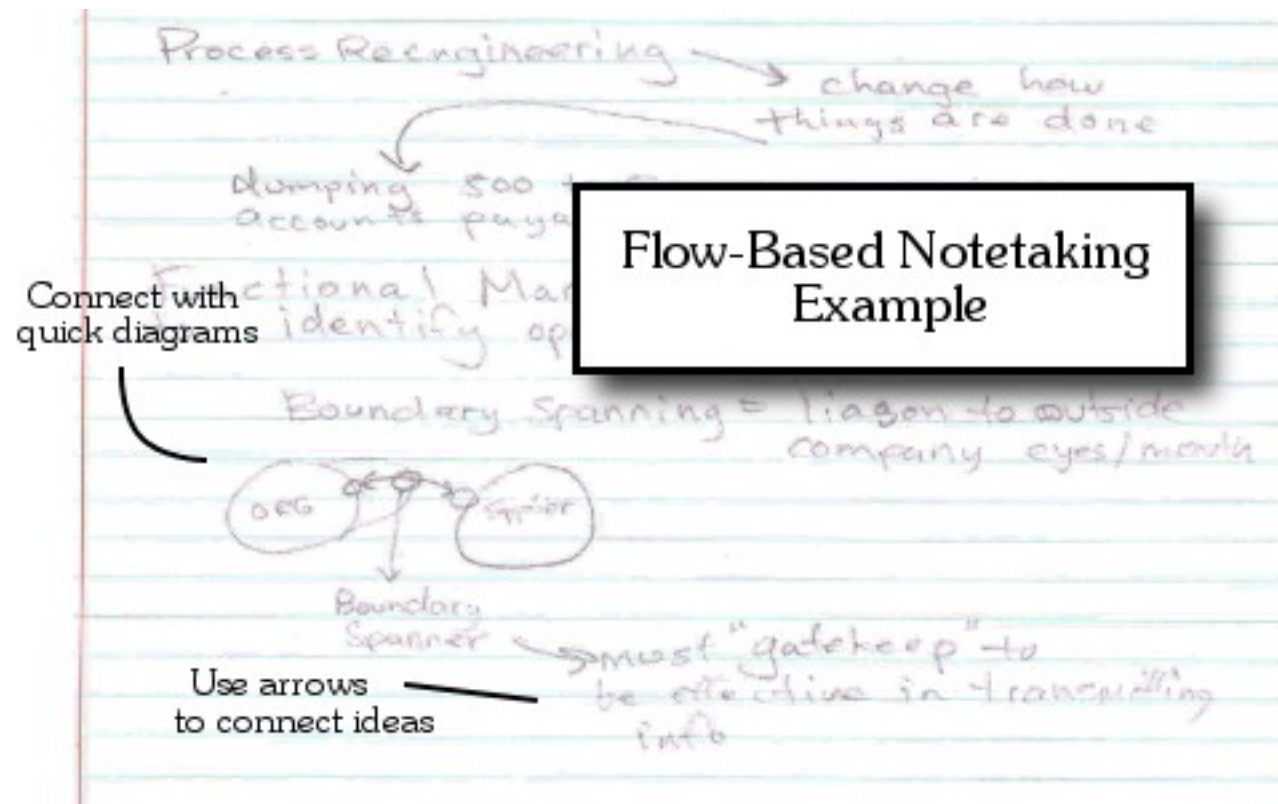
One technique I use during classes where there is a lot of information is flow-based notetaking. The goal with flow-based notetaking is providing a surface for connecting and linking ideas as they are reaching you. The linear, bullet-point style of notes that most people use is replaced with a much more fluid (although messier) format.

With flow-based note-taking you start by only writing out the major ideas. This means using a few words at most instead of entire sentences. This can reduce readability later, but it enhances learning during the lecture. Facts, dates, details and descriptions are reduced to just a few words, not lengthy paragraphs.

Once you get an idea written down, your next step is drawing a few arrows to connect it to other ideas. Instead of an ordered hierarchy of ideas, you want to represent the ideas as being interrelated components. This process more closely mirrors the actual holistic learning strategy, where ideas are linked into a web.

I tend to use flow-based notetaking as a method for using other techniques as well. Metaphor, diagraming and information compression are methods that can be used in conjunction with flow-based notetaking to enhance your understanding. This way you can write out major ideas and connect them to small pictures, diagrams or references to other subjects.

Remember that notes are only an intermediate step towards understanding. Having a beautiful set of perfectly written notes is useless if you don't understand the subject you are trying to learn. Flow-based notetaking, therefore is messier approach to taking notes, but one I believe is more effective at helping to understand the material.



Shadow Price

Every Model has both

Primal

&

Dual

Link Ideas to Images

gives results in terms of the amount of profit gained

gives results in terms of the value of constrained resources used to earn profit

Break Down Long Explanations Into Simple Diagrams

Canonical Form

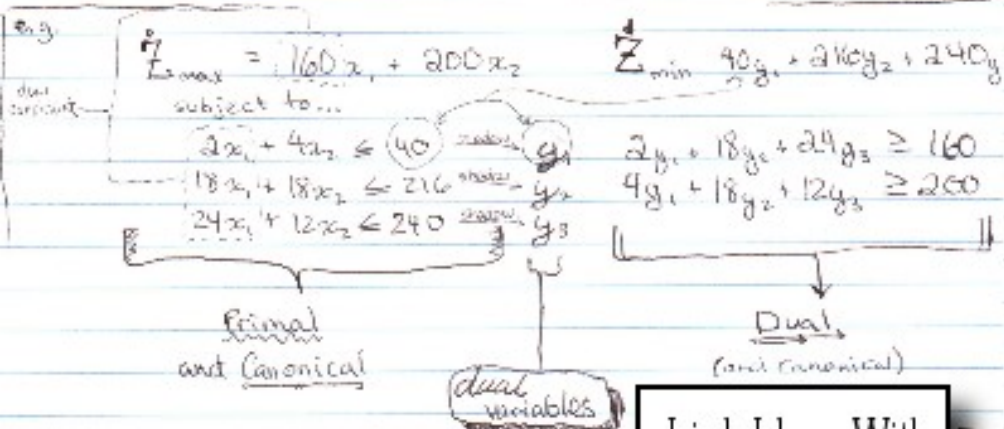


when maximizing

when minimizing

all $\leq$

$\geq$



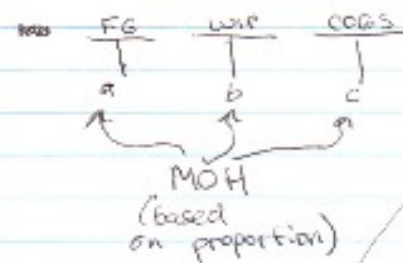
Link Ideas With Arrows

Here are a few examples of flow-based notes I took from two business courses. I took these during my class but you can retake your normal notes in a flow format after the class.

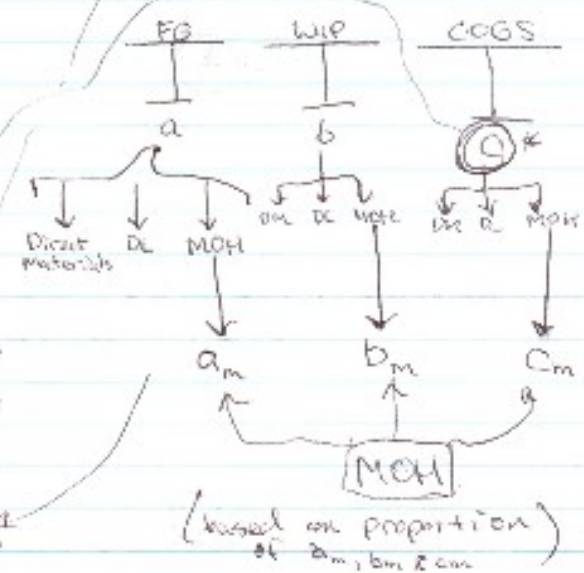
MOH balance

① Go straight to COGS $\rightarrow$ Easy, but inaccurate $\rightarrow$ use when balance isn't significant

② Apply to accounts based on ending balances



③ Apply to accounts based on the actual MOH used in each remaining account.



Remember not to include jobs that were manufactured in past periods

requires getting rates individual jobs

good when ratios of overhead vary from job to job

Hybrid Flow-Based Notetaking

Flow-based notetaking involves a trade-off between recording and exploration. With regular, linear notetaking, you can create an almost perfect record of what was said in a class. This method is useful if you need to review that information multiple times in order to learn it properly.

With flow-based notetaking you are sacrificing some later readability, for current understanding. By reducing the content of your notes and adding links or diagrams, the material can be learned more holistically. However, if the class has a high information density or you plan to review notes thoroughly later, there are hybrid strategies you can pursue.

Flow-Based Afternotes

The first hybrid strategy for flow-based notetaking is to take regular notes first and then recopy them into a flow-based format. If you are having trouble keeping up with the pace in a class, this strategy can give you more time to properly digest the information. Although it takes longer than a purely flow-based or linear notetaking style, it gives both readability and understanding.

I suggest starting with flow-based afternotes for the first month of trying this new notetaking style. This will ensure that you have a copy of your clearly organized notes in case you need to study them later.

Flow-Based Commenting

Some classes have an extremely high information density. When you are writing frantically just to get everything on paper, flow-based notetaking is almost impossible. Flow-based notetaking assumes that you can record all the critical information in a class in less time than it takes to teach. Most good teachers will give plenty explanation room and examples. During that time you can create the connections, metaphors and diagrams you need to learn holistically.

However, in cases when information density goes faster than you can record, flow-based commenting is an alternative strategy. Basically it involves writing down the key information and inserting links into your notes when there is a break. If a professor puts up a few dozen formulas you need to record, you could write all these down first. Following that, you could add more connections when the professor starts giving examples of how the formulas are used.

Recognizing Critical Information

The key ability with flow-based notetaking is to know what is important. What is the core information taught here? If you write down everything said in a lecture with equal emphasis, then you'll spend your entire class transcribing instead of thinking. Instinctively writing down every word written on an overhead transparency or Powerpoint slide is useless if you don't actually think about what you are writing.

With flow-based notetaking I cut down the amount of information I transcribe and emphasize on connecting and sorting that information in a way I understand.



Quick Tip!

Get comfortable writing your ideas in a flow-based format, before trying to do it during class. Flow-based notetaking works best when you have already mastered the Diagraming and Metaphor techniques in the next section.



Mind Challenge!

The purpose of this mind challenge is to get practice with flow-based notetaking, before deciding whether to use it instead of your current method for taking notes.

- 1) Buy a separate note-book and write “Flow-Based Notetaking Practice” at the top.
- 2) Commit for two weeks that once per day, you will take your existing notes (written during a class) and transcribe them in a flow-based style. That means taking ideas and writing them down, drawing connections between different ideas.
- 3) As additional practice, you may want to implement some of the other techniques into your flow-based notetaking style.

Linking Ideas

After you acquire information, the next step is to understand and explore those ideas. Understanding only the surface of information, often isn't enough to make it stick. Going beyond that can take a bit of extra time. But hopefully, as some of these methods become habits, you can use them automatically when you need to.

These techniques aren't necessary for every piece of information you learn. Many of the ideas I tackle in a class, I don't go through the entire formal process of finding metaphors, visceralizing or creating a diagram. Attempting to do so would mean all my time was wasted studying.

Instead, I use these methods selectively on information that falls into two categories:

- 1) Difficult information
- 2) Critical information

Difficult information could be a list of dates, or a series of ten steps which don't logically follow each other. Difficult information may be abstract, arbitrary, or simply be from a subject that you don't have many models to draw from. In these cases, applying linking methods can ensure those ideas aren't forgotten.

Critical information is like the foundation of a building. It forms the base from which you construct many other ideas from. A beginner learning about matrices would find that determinants and row reduction are both extremely crucial topics. Failing to understand these ideas (even slightly) could ruin any possibility of learning hundreds of other ideas from which they are based.

Information that isn't particularly crucial or difficult, doesn't warrant special techniques. Since much of the learning process is subconscious, you will probably form enough associations and models for it to stick without the intervention of a special technique.

Using Linking Methods to Form Models

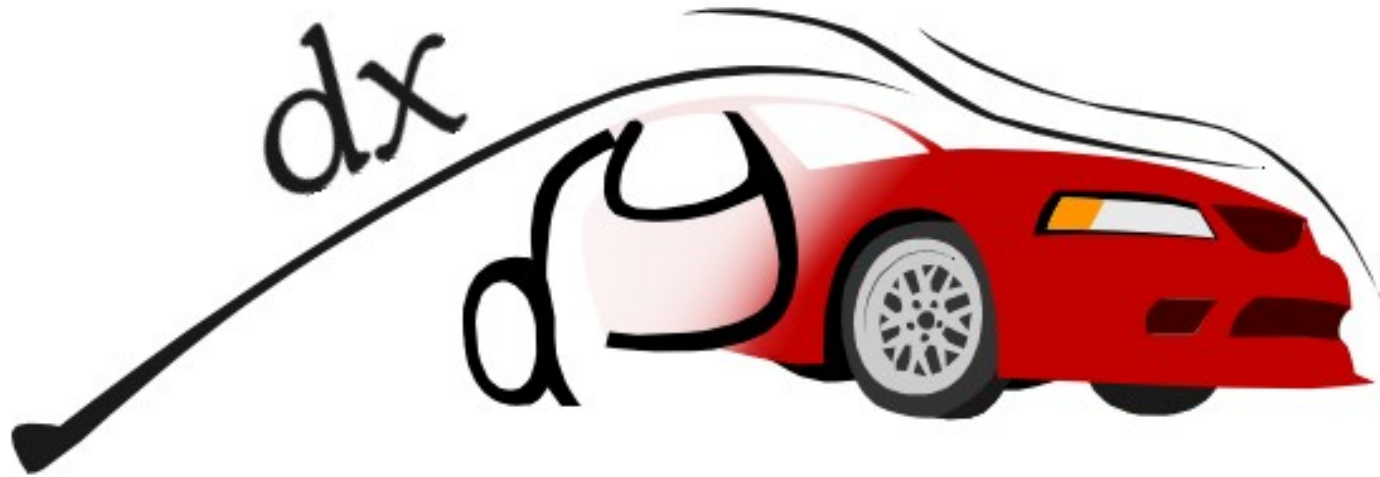
Earlier in the book I pointed out how models formed the basis of constructs. Simplified images and concepts that can be used to describe new information. These

linking methods are the primary ways to create models in holistic learning. Most of the time I create a model, I would use either metaphor, visceralization or diagrams.

Practice, Practice, *Practice!*

Like all the methods in this book, these techniques require practice to use effectively. If you aren't used to using them it can take 2-3 times as long to apply them. Furthermore, when you are beginning with these techniques, I suggest making use of a pen and paper. I don't write out metaphors and diagrams if I can imagine them clearly, but if you aren't used to these techniques, paper is the best way to practice.

Metaphor



Metaphor

Metaphors are a literary tool that describe an object by relating it to something that it isn't actually associated with. Saying a woman has an hourglass figure, doesn't mean that she is made of glass and sand falls from the top of her body to a lower half. Instead, it is a metaphor comparing the curves of her figure to the curves of an hourglass.

Similes are another tool in literature which are almost identical to metaphors. Similes involve comparing two unrelated things with "like" or "as". The man was built like an ox. My grandfather is wise as an owl. The wind felt like frozen knives piercing my skin. All of these are similes.

Metaphors work in stories by stealing common experiences and attaching them to uncommon ones. If a story said that the character John "had a bulldog face" then you could easily visualize him. Here the metaphor serves its purpose by linking the character John, who you have no mental basis to draw a description from, and relates it to a bulldog, which you have probably seen before.

The purpose of metaphors in holistic learning is similar. You want to create a bridge between an unfamiliar idea and a familiar one. While metaphors in literature are often used to provide visual descriptions, metaphors in holistic learning are more often used to connect similar processes, events or ordering of information.

For example, let's say you were in a psychology lecture learning about classical conditioning. Classical conditioning is the process of associating a stimulus and a response and was made famous by Pavlov's dogs. Pavlov noticed that by conditioning the dogs to associate a ringing bell with the arrival of food, the dogs would salivate when he rang a bell. After several times, the dogs would salivate when hearing a bell, even if he didn't bring food.

How could you use metaphor to learn this? The first step in metaphor is to look for something in your experience that models this process. As I grew up with Canadian winters, the first metaphor I imagined was walking through snow.



When you first walk through snow, every possible path is equal because they are all densely filled with powder. But after several walks through the snow, the first path you chose will become easier to walk through. This is because the compacting of snow under your feet creates a trail. Soon it is far easier to walk through one path than any other.

I can link this concept of walking through snow to classical conditioning by seeing the associations in the dogs brain. Initially, the bell could cause the dog to salivate or not (representing the expanse of snow with no trails). But after conditioning the bell with the arrival of food, the path from bell to food inevitably creates a trail through the snow. Eventually the dog will drool at the sound of the bell because that path has been so strongly conditioned.

Like most metaphors this one isn't perfect, but it can be a useful example. Coming up with a metaphor is a matter of following three simple steps:

- 1) Identify the information you want to better understand or remember. In our case it was classical conditioning.

- 2) Find something in your experience that matches part of the idea you want to understand. Perfect matches are often impossible, so compromise with a couple imperfect metaphors instead of a complete match. In our case we used walking through snow as an example.

- 3) Repeat this process and check for circumstances where the metaphor doesn't apply. With this example, walking through snow is a linear process, whereas brain neural connections have many different impulses running at the same time.

Sometimes a metaphor doesn't easily drop into your lap and requires more creative effort. Our snow-walking example fit snugly within the idea of classical conditioning. But, often metaphors require more effort in constructing.

Let's say you were taking basic calculus and needed to understand derivative. A derivative is the result of differentiating a function and has many useful properties in mathematics.

A derivative will measure the slope at any part of the parent function. So if you have a function that rises in a straight line upwards, the derivative will be flat as the slope is the same throughout the parent function. In a curved line, the derivative will have a shape that models how the slope changes at every position on the parent function.

The problem with this explanation of a derivative is that it might be hard to remember what a derivative represents. Creating a metaphor can help because it will connect the principles of differentiation to a common experience.

A metaphor you might come up with is driving a car. On the dashboard you have the odometer and speedometer. The odometer measures how far your vehicle has traveled and the speedometer measures how fast you are going. If you graphed your odometer and speedometer over time, the speedometer should be the derivative of the odometer. The slope of your position graph would be the speed graph.

Tips for Improving Your Metaphors

The problem with both of the examples in the last section is that they are based on previous metaphors I've already constructed. When you are given an idea, you might have few starting points in creating a metaphor. How can you improve the speed and quality of the metaphors you make?

1) Start by asking for a metaphor.

Unless you ask yourself what a good metaphor would be to describe an idea, you probably won't find one. Keep the concept of metaphors in mind the next time you encounter an idea you have difficulty relating to.

2) Pick the first thing that comes to mind.

Finding metaphors is a process of creative experimentation. This means you'll probably have to go through several incorrect metaphors before finding one that seems to model your topic. I can usually go through 4-6 incorrect metaphors in a minute or two before finding one that seems to work.

My suggestion is to not inhibit the creative process by claiming you can't find a good metaphor. Instead, just grab the first thing that comes to mind. Look for where it fits and where it doesn't fit. If it doesn't fit, discard part of it and try again.

3) Refine and test your metaphors.

Often you'll find a metaphor that works as a description for some parts of your idea, but not all. My suggestion then is to collect multiple metaphors that each explore the idea from a different angle. Not only does this reduce the chance of making an error in understanding, but it gives you a greater amount of links to a single idea, which improves the quality of your construct.



Mind Challenge!

The purpose of this Mind Challenge is to improve your ability with using metaphors to understand and remember abstract ideas:

- 1) Commit for at least two weeks to go over a subject you are studying once each day. After you read your textbook or notes, write out at least 3-5 major ideas.
- 2) With each idea, write out a possible metaphor that could be used to explain it.
- 3) If the metaphor isn't complete, try looking for one or two more metaphors that could also associate to the same idea.
- 4) Use the 10-Year Old Rule. Could you explain this metaphor to a ten year old child? If your metaphor is just as confusing as the original example, repeat it to yourself and rewrite it until you get something that is simple and easy to understand.
- 5) Repeat this process with the other 2-4 ideas in your textbook until you are satisfied that you understand them.

Visceralization



Visceralization (Visceral + Visualization)

You've probably heard of visualization. This is the process of creating a mental image. Visceralization is my word to describe imagining not only a mental picture, but sounds, sensations and emotions. Often a mental image will work, but connecting an idea to several senses and even emotional states can create a stronger link than a picture.

To visceralize an idea, the best way to start is simply with forming a mental picture. This works best when the idea is already fairly easy to imagine. Imagining the process of light entering your eyes, traveling through rods and cones in your retina, through your optic nerve and into your brain is a process easier to visualize than, let's say, different philosophies of ethics.

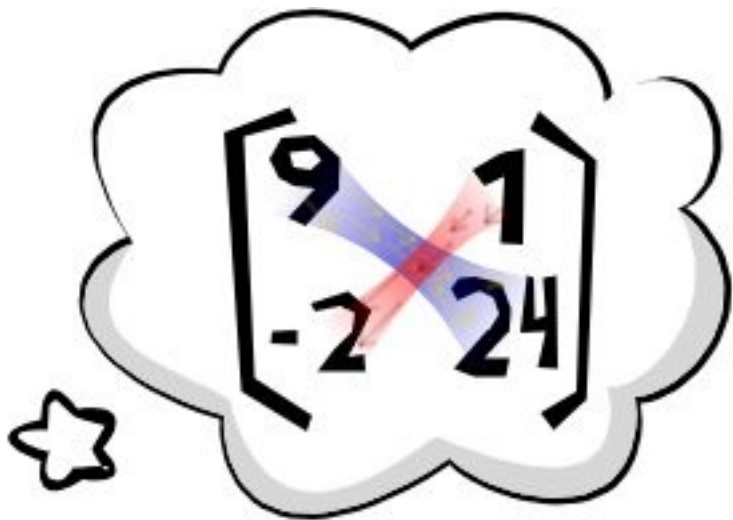
Visceralization works best with concrete information. I've found abstract information to be better suited for metaphors, although there is considerable gray area between the two.

After you have formed a mental picture, begin adding other sensations into the image. If you are trying to visualize the perception of light, imagine feeling the light enter your eyes. You might want to visualize a specific sound as it hits your retina and surges down the optic nerve.

When I started to learn about matrices, I had to learn how to calculate a

determinant. A determinant is found by multiplying the top left and bottom right numbers and subtracting from that the product of the top right and bottom left numbers. This abstract process would have been difficult to remember if I hadn't visceralized it first.

Visceralizing this process of calculating a determinant involved first imagining a 2x2 grid containing four numbers. I then imagined my right hand swiping through the top left and bottom right numbers. As I did this I imagined a blue streak and the feeling of added weight being formed in my hand. Then I imagined my left hand swiping through the top right and bottom left numbers. From that I imagined a red streak and weight being reduced in my right hand.



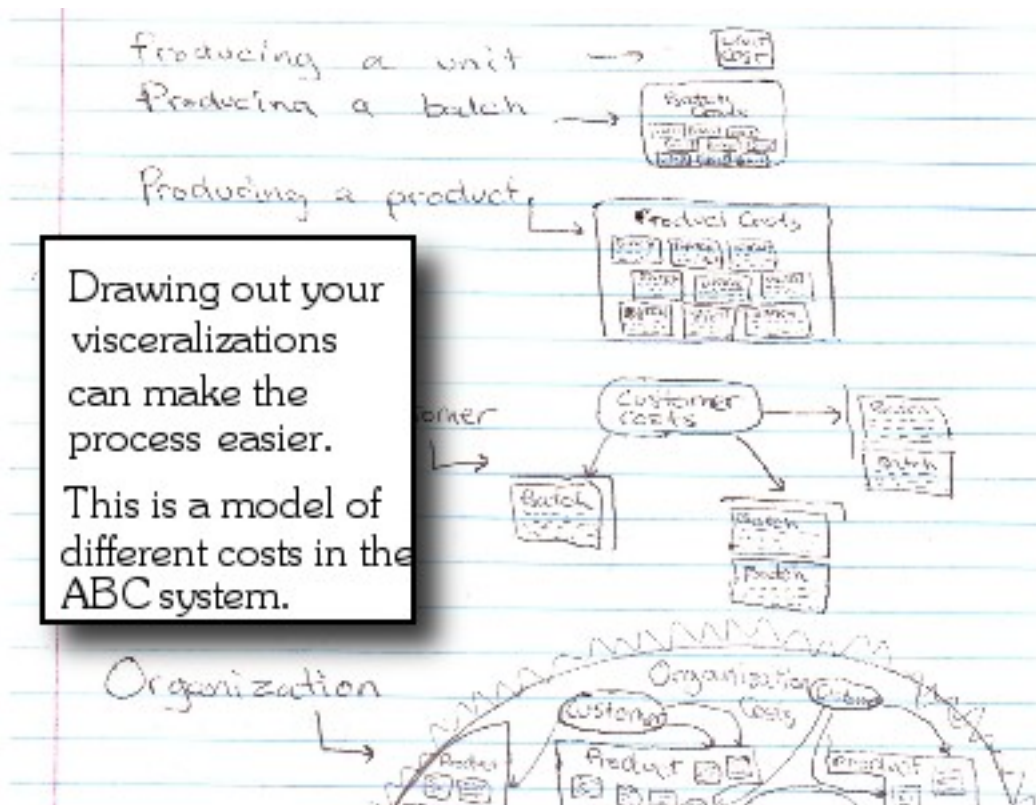
By connecting specific senses, sounds and kinesthetic actions to the original idea, it was easy to remember for future questions.

Some learning experts claim that there are three main types of learning styles present in people: visual, auditory and kinesthetic. Visual learners (I presume I'm a highly visual learner) require information to be presented in pictures to understand them. Auditory

listeners need to hear information in terms of sounds and instructions to understand. Kinesthetic learners need to feel physically involved by touching and manipulating in order to understand.

While I don't think labeling yourself as one learner or another is entirely constructive, it is useful to see that different people learn differently. Visceralization is an attempt to take information that isn't in your preferred format and translate it into a style that you can easily understand and relate with.

An added step to visceralization is giving ideas emotional impact. It is well-known by psychologists that memory is influenced by being in different emotional states. Connecting ideas to emotional impacts (even small ones) can make them more memorable than if they are completely dry.



How to Visceralize

1) **Identify the concept you want to visceralize.** This could be a process in biology, a function in computer programming or mathematical concept in calculus.

2) **Start by picking a mental image to base the idea from.** If you aren't used to visceralization, I suggest making a crude drawing of the idea on paper so it will be easier to hold for the duration of this exercise.

3) **Does the process move through time, or is it a static image?** Taking a determinant has several steps, so it flows through time. If your concept does flow through time, try to visualize it moving as if you were watching a movie.

4) **Now add another sensation to your image.** Try picking it up, acting upon it or feeling it as a sensation within your body. Link the sensation to the movement or feel of the mental image.

5) **Add other sensations or emotional impacts to the image.**

6) **Refine and repeat the image until you can bring it up in just a few seconds of thinking of it.**

Like metaphor, visceralization is a process of creative experimentation. You may not be able to come up with a great picture at first, so just pick something to get you started. You can always refine and discard parts of the picture that don't work until you get an image that is memorable and vivid.

I strongly suggest making the use of paper and pen when practicing this technique. Using paper can help you get practice with forming memorable images.

Metaphorisceralization

Alright, so perhaps blending three words together (metaphor, visceral and visualization) is a bit much. But metaphors can be combined with visceralization for abstract but important ideas you want to better understand and remember. The process is exactly the same as visceralization, except you use a metaphor to find a concrete description of your abstract idea first.

With our calculus example from the last section, this could mean you visceralize driving your car while watching the speedometer and odometer moving. Sometimes adding a visceralization to metaphor is overkill, but it can be helpful for tricky concepts that are hard to describe and imagine.

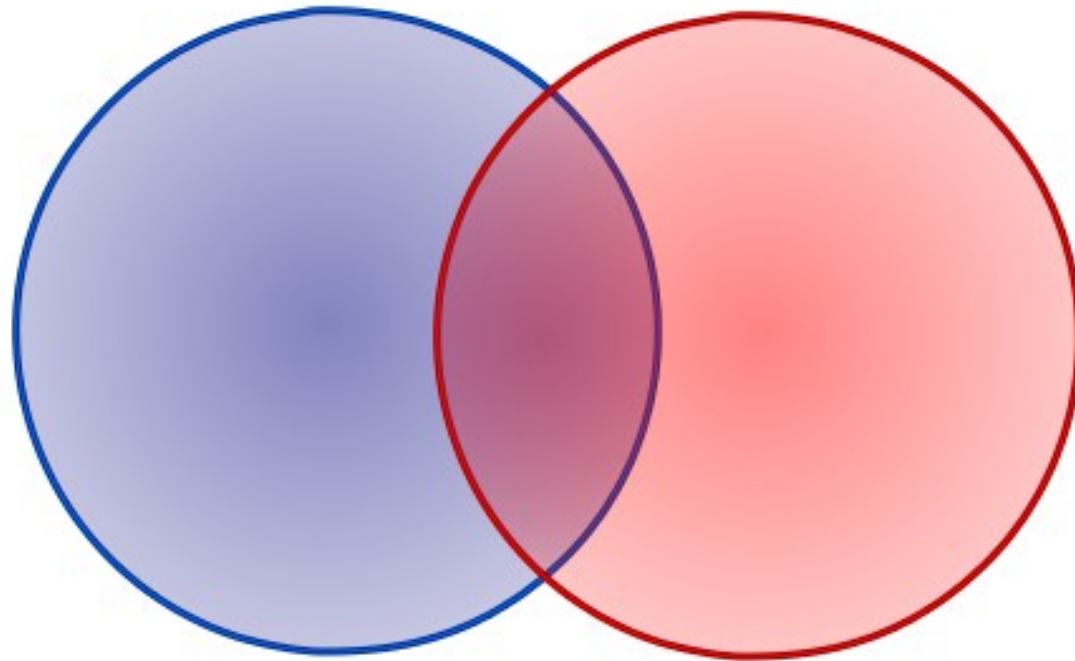


Mind Challenge!

The purpose of this mind challenge is to give you better flexibility and skill with your ability to form mental pictures and link in different sensations.

- 1) Get a few blank sheets of paper and a pencil for drawing.
- 2) Pull open your textbook and pick a few key concepts or ideas. If you are just starting with visceralization, pick ideas that are easy for you to form mental images. Completely abstract ideas or incredibly complex ones might be too hard to start with.
- 3) Form a mental image, add sensations and emotions to your idea as the steps are for visceralization on the previous pages.
- 4) Quickly sketch a picture of the idea based on your mental image. (30-60 seconds max.)
- 5) Repeat this with several other ideas each day for a two weeks.

Diagramming



Diagraming

The final technique I have for exploring ideas is diagraming. Diagraming is a simpler form of visceralization. Creating diagrams takes more time than visualizing an image, but it is easier to do and can be used on fairly abstract ideas that would be hard to imagine otherwise. Diagraming can also be combined easily with flow-based notetaking and active reading while you are learning.

A diagram is a picture that compresses several ideas together in an easy source. The most common types of diagrams are charts. A scatterplot can compress hundreds of individual data points into a single graph. Flow-based diagrams can chart a complex series of interactions and steps onto one image that is easy to see.

Often books will come with premade diagrams to simplify the information they contain. Where they lack diagrams, constructing your own can be useful to chart the flow of ideas, picture different concepts or link different concepts together.

There are three main types of diagrams I'll cover here: flow, concept and image diagrams. Using some combination of these three types can help you understand almost any concept. The only disadvantage of diagraming is that it takes longer than metaphor or visceralization. However, with practice I've been able to speed up this process as well, so that I can quickly create a crude diagram in less than a minute.

Flow-Based Diagrams

A flow based diagram is useful for:

- Charting a sequence of steps. (How to do long division, the process of creating a cash-flow statement, etc.)
- Charting historical events, creating branches linking events together not only through causation but through the time period they rest.
- Mapping out a system. (e.g. How a function works in a computer program)

The basis of a flow-based diagram is that you start with a single element, and draw connecting arrows to different ideas as they relate to that element. I could have created a diagram for the steps I undertook to write this book. Starting with the initial idea, I could have drawn lines showing subsequent steps, branching off where there is more options.

The importance in a diagram must be for your understanding, not creating a pretty picture. It is easy to get caught up in making a tidy and beautiful picture and waste hours on a diagram which could have been built in just a few minutes. Simplify and be messy if you need to.

Concept-Based Diagrams

A concept based diagram links together ideas and is closely related to flow-based notetaking. Here the associations aren't necessarily different steps in a sequence or an order of dates, but relationships between ideas. Create arrows connecting different ideas, but include 1-3 word snippets on each arrow to note how the two ideas are related.

If I wanted to create a diagram of different financial accounting principles, I might start by writing down the four different statements: balance sheet, income, retained earnings and cash flow. From that I could then connect ideas together to form a web of information.

If I wanted to map out the characters in a novel, I could start with the protagonist in the center and map out different characters and relationships that character had. Mapping of this kind can help organize your thinking on a large quantity of information.

Start with the most important ideas and branch off into details and minor ideas.

Image Diagrams

Not technically a diagram, but image diagrams are important for diagraming so I'll include these here. An image diagram is any rough sketch or doodle used to represent an idea or the association of ideas. Here you can create a small image either by itself or within larger diagrams to make them more memorable.

I frequently make small images in my notes or diagrams as a reference for important ideas and to make them more clear inside my head. Drawing detailed images can be a waste of time if it takes too long, but a simple 10-20 second doodle can be valuable if it helps connect the ideas properly.

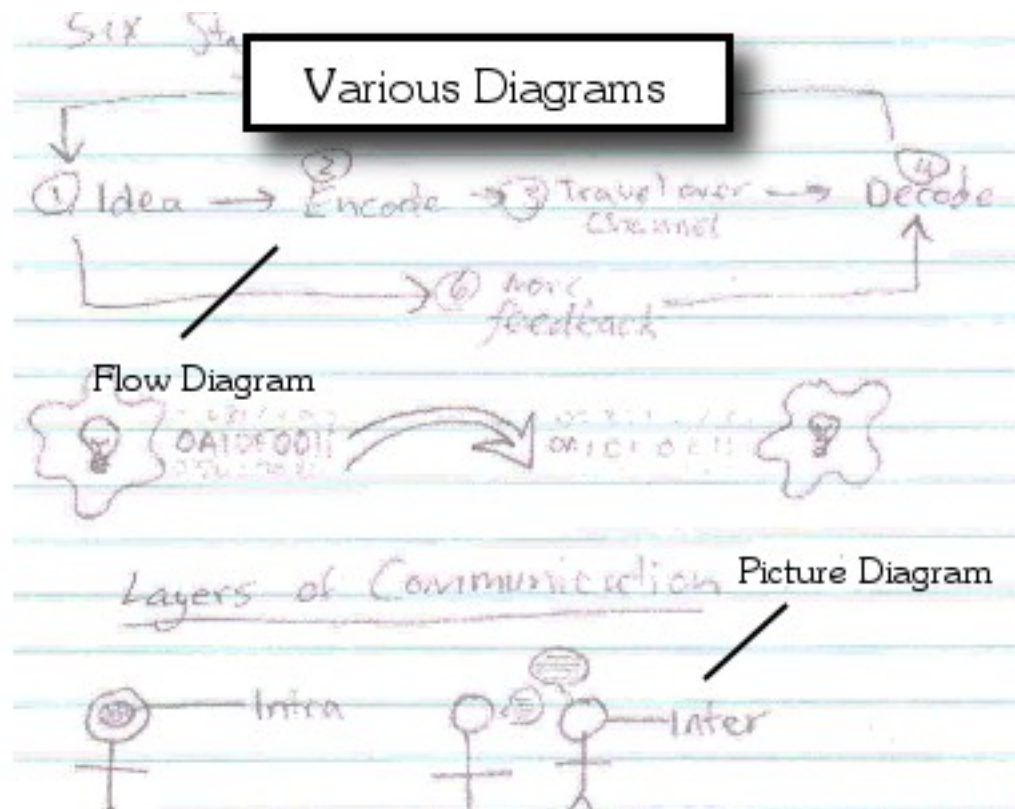
Combining Diagrams with Metaphor and Visceralization

Diagrams can be mixed with the usage of metaphor and visceralization to improve your understanding of ideas. After you have found a metaphor or mental image to describe an idea, create a quick sketch, connecting different concepts together on paper.

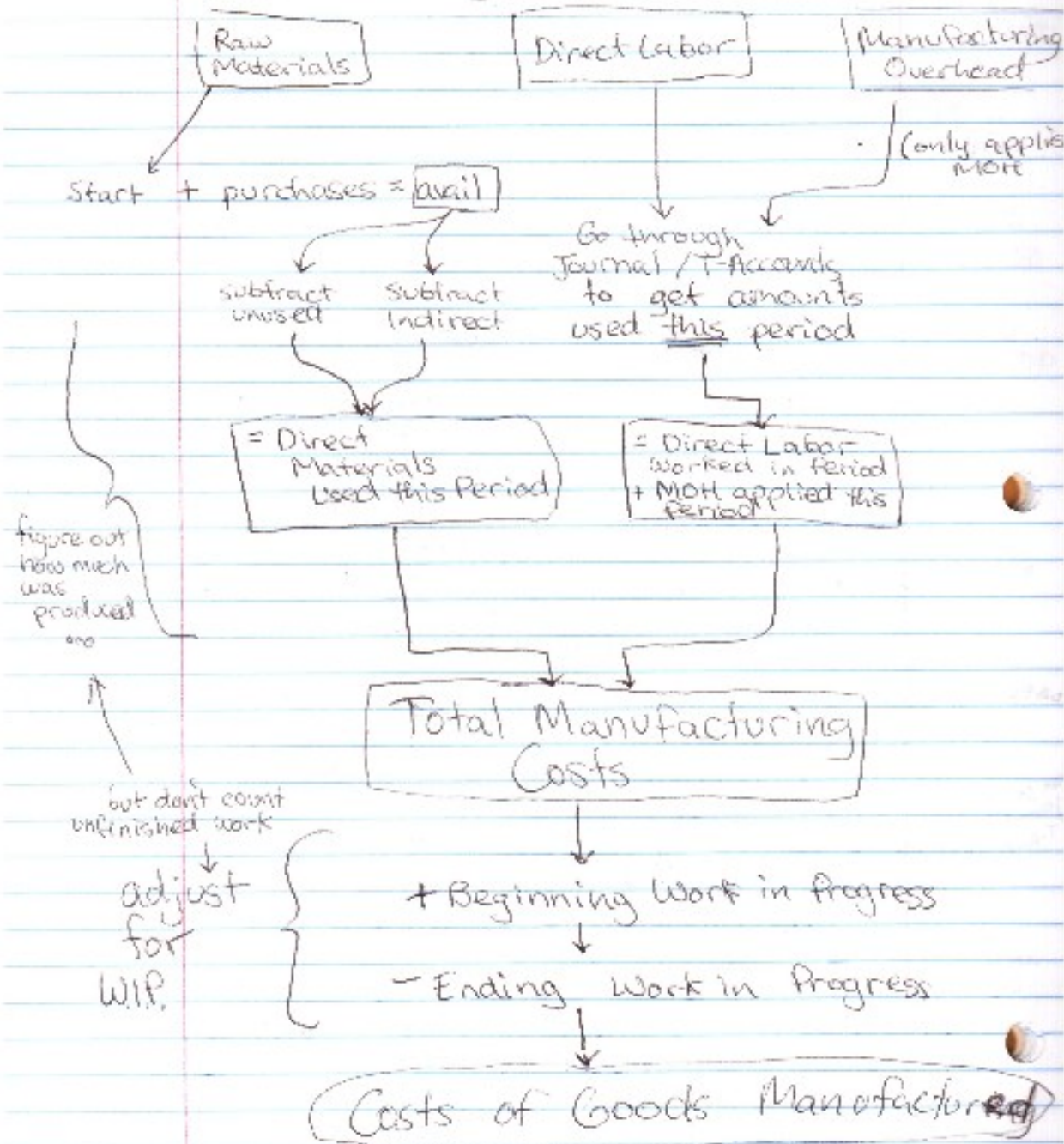
With our calculus example from the previous section on metaphors, I could have drawn an image of a racecar and linked the odometer to a position graph and the

speedometer to a speed graph. I could have associated the two of those graphs with a symbol for differentiation making the relationship between all elements clear.

Adding a diagram to a metaphor or visceralized image may be unnecessary for many situations, but it can be helpful for remembering a complex idea.



Finding COGM



Flow Diagram

Here is a diagram I made during a class in accounting.

Diagrams can make it easier to visualize abstract rules and relationships.



Mind Challenge!

The purpose of this challenge is to enhance your skill at making diagrams and increase your diagraming speed. Since diagraming is the slowest of the three linking techniques mentioned in this section, improving speed is important if you want it to become a useful technique for your daily studying without it becoming a burden on your time.

- 1) Get a sheet of paper and find an idea or set of concepts that could be made into a diagram.
- 2) Start a timer for no more than 2-5 minutes and begin creating a diagram using one or a combination of the different diagram types.
- 3) When the timer beeps, stop what you are doing and make a note of the amount

of information you've managed to store, the clarity of that information and how many different relationships or connections you were able to make.

4) Repeat this exercise once a day for two weeks until you're comfortable making quick diagrams of several different ideas.



Quick Tip!

Once you've mastered diagramming, try combining it with flow-based notetaking in classes or active reading for your textbooks.

Handling the Arbitrary

Arbitrary information can present a challenge when trying to learn holistically. Just how do you form links, metaphors or visceralizations with things such as:

- A list of dates.
- The scientific names of different parts of your anatomy.
- A series of steps.
- Rules and laws that have no clear basis, but still need to be followed.
- Scientific formulas that take hours to derive manually.

Normally these are situations that require memorization. Holistic learning strives to avoid rote memorization whenever possible. Instead, using metaphor, visceralization and diagramming to create models and eventually constructs is a better and faster way to use your brain. Unfortunately there are some cases where these tools won't work.

Information that is arbitrary, or too large and complex to understand fully requires different techniques. If you find that using the linking methods won't help you understand the material, or it simply takes too long for the amount of information you need to learn, arbitrary methods can come in handy.

These arbitrary methods hover the line between rote memorization and holistic learning. The link, peg and information compression methods may be seen as just another set of memorization techniques. However, they have been designed so that memorization can be done quickly and work the way the brain is set up to learn. If rote memorization is a blunt club and holistic learning is a scalpel, these techniques would fall somewhere around the range of a steak knife.

Even more so than the linking techniques, the arbitrary techniques require practice. There are two reasons more practice is required with these methods:

- 1) *You should use them infrequently.* I use the link and peg techniques sparingly, since the linking methods described in the last chapter produce a deeper understanding. Because they aren't being used constantly, practice is important to keep these tools sharp.
- 2) *They are more complex than linking methods.* Link, peg and information

compression are less intuitive than the linking methods because they use a bit of mental trickery to get the job done. Because they are less intuitive, they require more practice to pull off successfully.



Quick Tip!

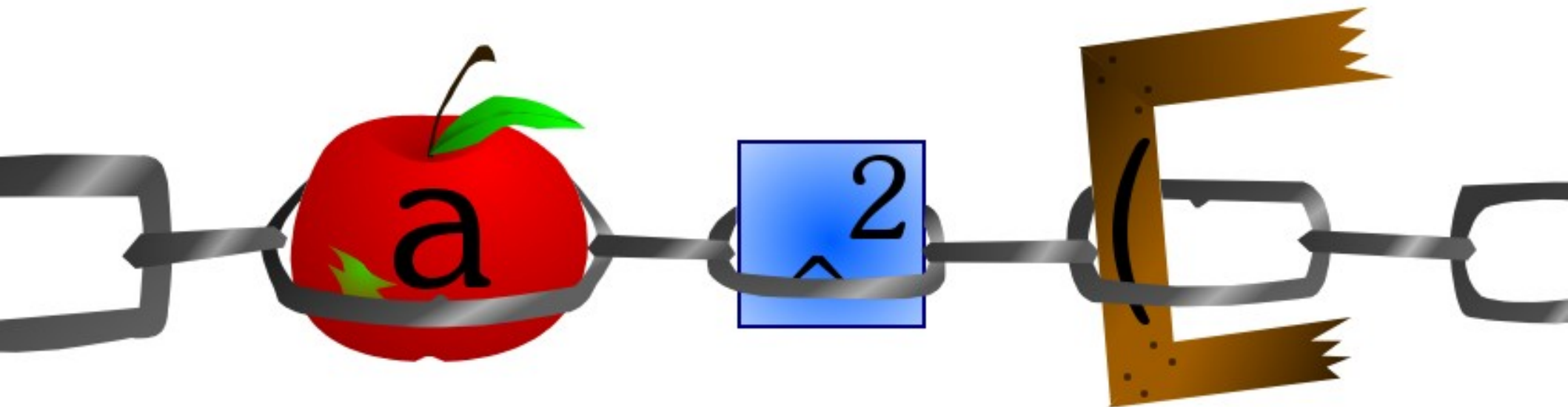
Arbitrary information has a weaker structure. Always try to shift information into a layer that is easier to learn:

Weak – Learning through repetition.

Better – Learning with linking, pegging or information compression.

Best – Learning with metaphor, visceralization or diagrams.

The Link Method



The Link Method

I didn't create the link method, as it is a memory trick that has been around for years. The link method is called that because it serves the purpose of linking a series of ideas together, like a chain. As long as you can access one link on the chain, you should be able to easily travel to any other link on the chain.

The link method works best for storing a sequence of arbitrary information. Steps in a process, sets of names or formulas written out in sequence all can be tackled using the link method. As you get practice with the link method you can chain together a series of several dozen ideas with only a few minutes of focus.



Quick Tip!

Despite the name (I didn't invent the technique) the Link Method has less to do with holistic learning and sits closer towards rote-memorization. It can be a useful method, but I would stick with metaphor, visceralization and diagraming where possible.

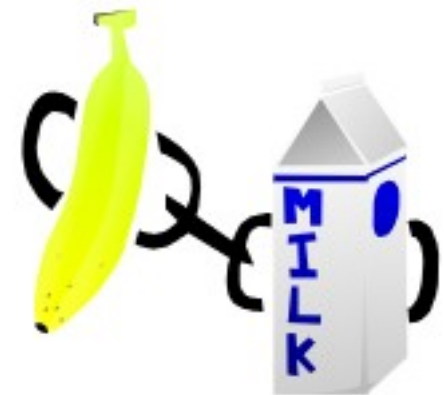
Steps to Use the Link Method

Step One: Create Your Sequence

Start by writing out on paper a sequence of information you want to store. Break apart the information into units you can immediately understand. It is important that the information be displayed in a linear list because that is the only way the link method can hold data.

A simple example might be linking together a grocery list:

- 1) Bananas
- 2) Milk



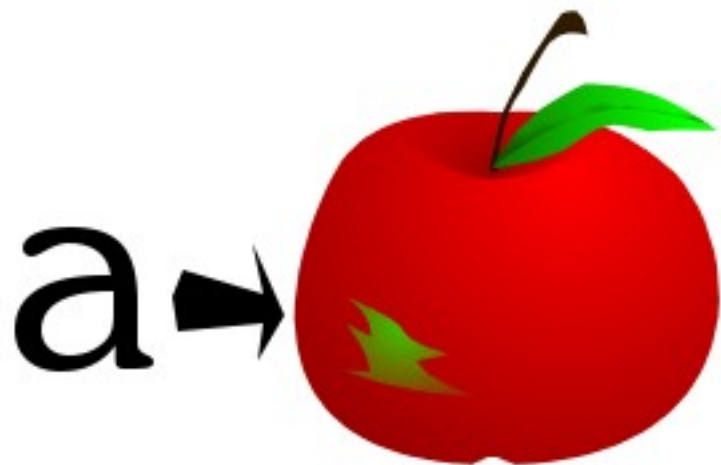
- 3) Baked Beans
- 4) Butter
- 5) Jam

While I'm sure few classes would get you to remember a grocery list, the principle remains the same.

Turning a formula into a sequence requires that you convert it into a form you might enter into a calculator. Therefore a formula such as $V = 1 + x/b$ could be written as V = 1 PLUS X DIVIDE B. V, =, 1, PLUS, X, DIVIDE and B would be your elements in the series. Complex formulas may take time to break down if you aren't used to this approach, but it can still be helpful if you are worried you might forget.

Step Two: Symbolize Each Object in the List

When dealing with an abstract list, you need to create an easily recognizable symbol for each item in the list. In a management course, I needed to remember Henry Fayol's ten principles of management. I used the link technique and created a symbol for each element. Division of labor became a knife and remuneration of



personnel became a carrot on a stick.

It is important that the symbol immediately helps you conjure the original concept. Picking symbols that don't give any significance to the original idea aren't useful, because the link method can only store visual symbols, not abstract ideas. Unless you feel confident you can convert your symbol back

to the original idea, you can't use the link method.

With formulas, you can memorize elements of the formula by creating an object that starts with the same letter as the algebraic symbol and coming up with a shorthand for different mathematical symbols. The variable **a** in a formula could be symbolized by an apple and raising a term to the power of two could be visualized as a colored square.

Write out your symbols next to each item in your sequence. As you get better with linking, you might not need to write all these steps. When you start with the method, write them out so they become habits.

Step Three: Create Your Links

This step is the basis of the link method. What you want to do is create vivid and exaggerated mental images that associate two symbols in the sequence. This means you will need to create a mental image linking 1 and 2, 2 and 3, 3 and 4, etc.

With our grocery list we would have to first create a mental image that links bananas to milk. For this I might imagine a giant banana that has cow spots on it. I could also imagine a farmer milking a banana's udders instead of a cow.

These ridiculous mental images are important for creating the link. If the link isn't wildly exaggerated and vivid, it will be difficult to recall. Simply picturing an ordinary jug of milk next to a banana wouldn't be sufficient to create a strong link. But a cow-spotted banana or a banana being milked are both crazy enough to stick in memory.

After you create link between two items, make a link between the next two items. So for our grocery list that means you would need a link between milk and baked beans. Again, a cow-spotted bean or a giant bean wrestling a carton



of milk might be vivid enough to create a link.

Try continuing with our grocery list to create your own links between beans and butter and butter with jam.

After you are done created a list, go through all of your links to see that one easily follows into the next. If you can't find the next link after viewing one, you probably didn't associate a vivid enough picture to the resulting details.



Quick Tip!

One attribute the Link method shares with holistic learning methods is the need for imagination. Don't be shy, trying to pick only “reasonable” connections between ideas. Linking works best when the mental pictures are exaggerated and bizarre. Holistic learning works best with many, varied connections.

Challenges With the Link Method

Although the link method can be more time consuming than other techniques, it is reliable and your speed will improve with practice. There are a few other challenges that can arise with using the link technique you will need to be careful of when creating a list:

Repetition of Symbols

Lets say you want to remember the following formula sequence:

$r, =, \sigma, (, Z, x, *, Z, y,), \text{Divide}, (, n, \text{Minus}, 1,)$

Overall this formula is a good candidate for the link technique. Unfortunately there is the repetition of the “(“ twice and the “)” twice. This means that you may inadvertently jump from the first “(“ to the n, instead of the first Z.

When symbols are repeating and you are worried about confusing the order, I suggest labeling each with a color, so that all elements in the sequence are unique. This way your link between “ σ ” and “(“ might use a blue bracket and the Division with “(“ might use a red bracket.

Broken Links

One failing of the link technique is that if one link in the sequence is broken, you can't retrieve any of the following links. This is why I suggest keeping the sequences short to about 5-15 links maximum. Beyond that it is better to chunk the entire sequence into a series of smaller links.

This way you might have a formula broken into three parts. Those three major sections would form a linked list of three and each major section would begin a new linked list containing 7 or 8 elements. This mental re-stitching can take extra time, however, so using it for every possible formula might not be advisable.

Indecipherable Symbols

As I mentioned earlier, the symbols need to have a clear description that is immediately obvious. If your symbols lack clarity, the linked list will fail. Some people who master the linking system begin to create their own shorthand, using symbols to represent common ideas, phrases or even syllables of words.

Lost Triggers

In some lists, you may want to add one additional link, a link between the first item and a trigger. With our grocery store example, how can you remember your linked list if you can't think of the first object on that list? In this case you might want to associate the first object on the list with a trigger, say your bags you take the grocery store, or the storefront itself. On seeing the trigger you should be reminded of your first link and all the links following it.

The same method can be done with information for tests. Linking the first item with an idea, say a formula with what it calculates or a list of principles with the theory they represent can be helpful in completing the linking process.



Mind Challenge!

Learning the link method will probably take more time to master than the methods of the previous section. Try committing for an entire month to practice using various lists from your textbook or notes.

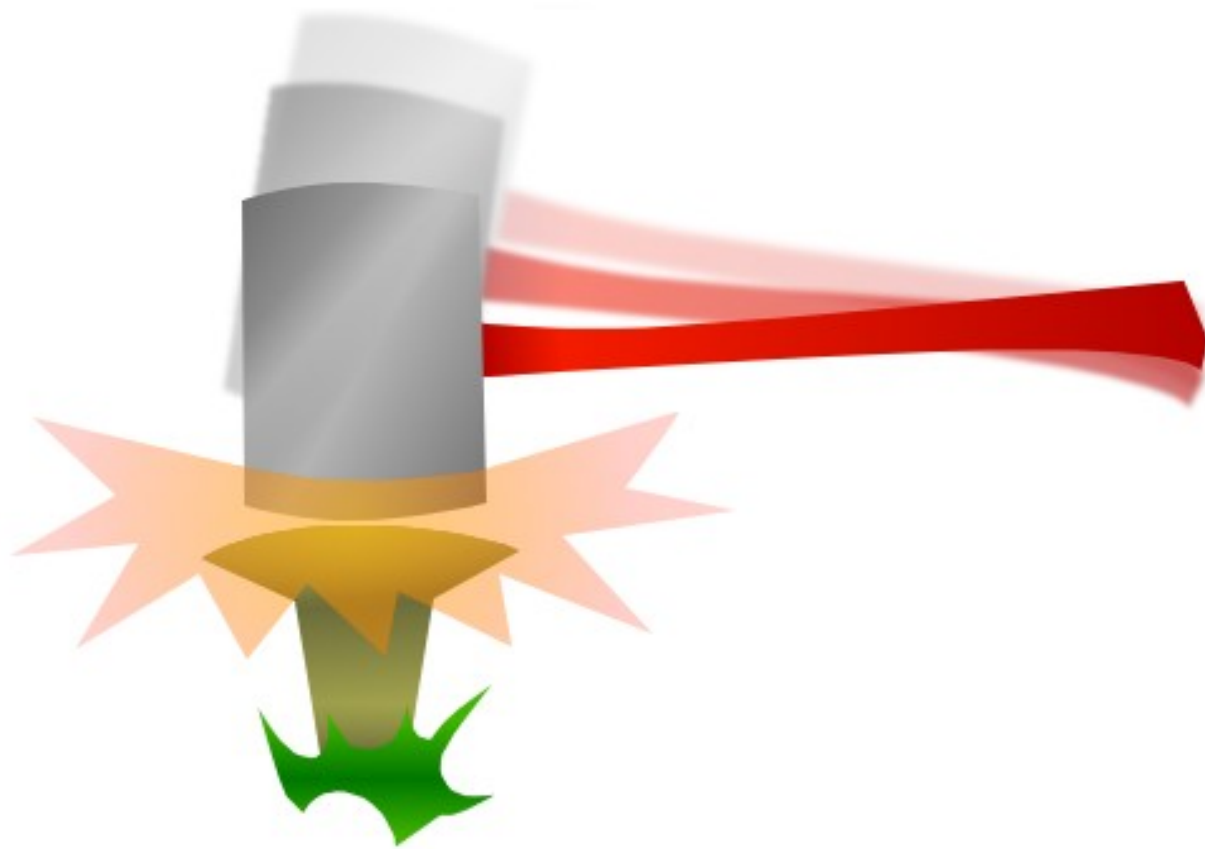
- 1) Pick a formula, list, group or any set of ideas that can be expressed in an ordered sequence.
- 2) Go through the above steps for creating a linked sequence, breaking it into a list, creating symbols and forming the proper links. Review your linked list to ensure all the ideas are vivid, repetition of symbols is minimal and the symbols are easily decipherable.

3) After your first week of the mind challenge, pick a random list from the previous week or weeks and attempt to go through it until you can reform the original formula, group or list.

4) After the second week you may want to time yourself to improve your speed at using this method.

I am not a perfect master of the link method as I prefer the holistic linking methods of metaphor, visceralization and diagraming to these techniques where possible. Since other books have been written about the link technique, I recommend doing a Google or Amazon search to find more information if you want to go beyond the basics of linking.

The Peg Method



The Peg Method

The peg method is similar to the link method, except it expands upon the basic idea. With the peg system, you store information attached to numbers, so you can recall any of the ideas without needing them to be in a specific sequence. The peg system can also be useful for linking together a series of numbers as in dates.

With the peg system you aren't linking a series of ideas with each other but with a specific numbered slot. I only use the peg system to record no more than a dozen items, however advanced versions of the peg system can store hundreds of items independent of any specific sequence. I'll discuss these methods briefly, but I've found being able to store up to 12 items by pegging to be sufficient for learning most ideas.

The peg system is based off of creating symbols for the basic numbers. My implementation of the peg system uses rhyming words to easily store the numbers 0-12. These can then be used as empty slots for storing concepts or steps in a process.

0 - Hero

1 - Gun

2 - Shoe

3 - Tree

- 4 - Door
- 5 - Hive
- 6 - Sticks
- 7 - Heaven
- 8 - Plate
- 9 - Wine
- 10 - Pen
- 11 - Ribbon (I know, not *quite* a rhyme with
- 12 - Oven (similar sounding to dozen)

Once you get the numbers 0-12, you can then associate any symbols that you need to peg to one of these thirteen slots. Forming a picture of a large bottle of wine fighting a knife might cause me to associate the 9th slot with a knife (my symbol for Henry Fayol's principle of division of labor).

With this sense you operate exactly as you did with the link system, except you



associate each item not with the one before and after it in the sequence, but with a specific position on the numbers 0-12. This way if one link becomes broken, every other link can still be recalled easily. The only effort required is remembering your 0-12 rhyming symbols.

Using the Peg Method to Store Dates

I haven't taken too many subjects that required excessive date memorization, but from my brief experience, I've found the peg system to work well in storing numbers. Let's say you wanted to link Columbus's discover of America to the date 1492.

The first step would be to create your trigger that could store the event for the date in question. In our case this might be an image of a ship crashing onto a crudely drawn picture of North America.

The next step would be to link the number 1, 4, 9 and then 2, to the date in question. So for us this would



mean first linking a picture of a gun to the trigger image. Perhaps an oversized pistol holding up poor ol' Christopher as he crashes into North America? Next you would need to link gun and door, door with wine and finally wine with shoe. Then as you play back the sequence of links you could get the numbers 1, 4, 9 and 2 or 1492!

Advanced Pegging

An advanced form of using the peg technique involves associating a phonetic sound to each digit 0-9, rather than a rhyming image. You can then use the phonetic sound to form a series of words to store dozens, if not hundreds of numbers in a sequence instead of just a couple. I won't go into this form of pegging, simply because I've found it to be unnecessary for more than party tricks.

However, if you want to impress your friends with your ability to remember a string of hundreds of numbers, you might want to do a search looking for more information on this advanced form of the peg technique.

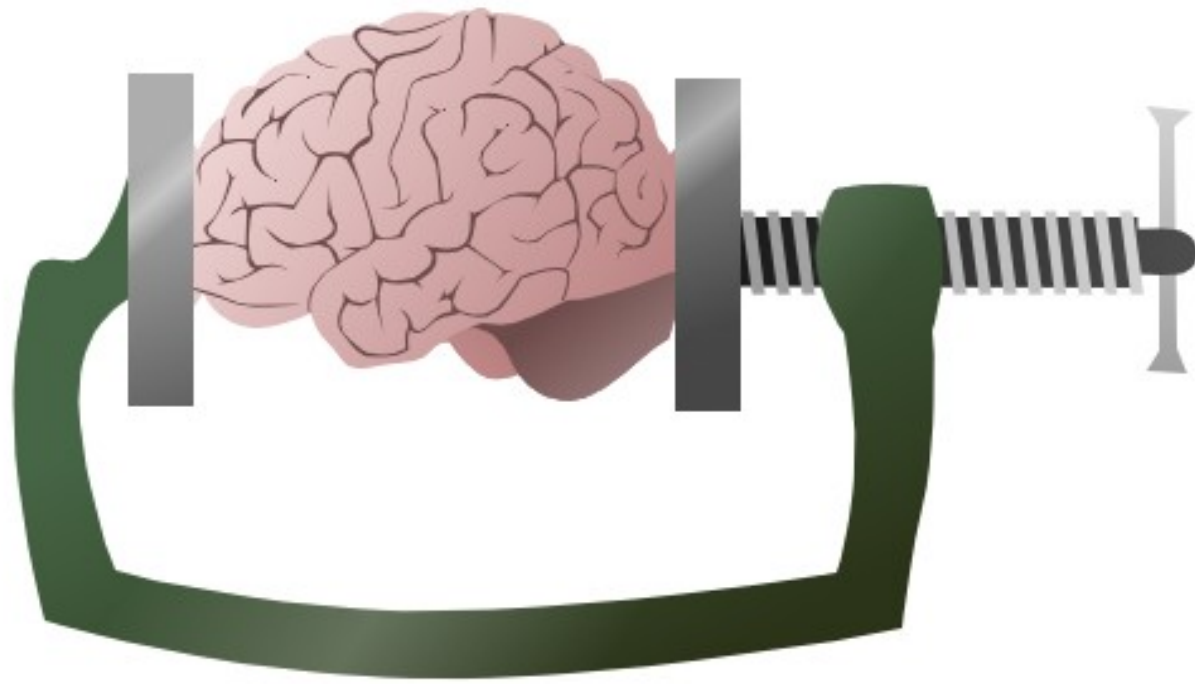


Mind Challenge!

The peg method is built off of the link method, so do not start this Mind Challenge until you've completed the previous one for the link method. Here the goal is to familiarize yourself with the peg method and begin using it to store pegged lists of information and series of numeric information such as dates.

- 1) Pick a list, group or date suitable for using the link method. Make sure it has no more than 13 items if you are trying to store a pegged list.
- 2) Follow the steps for pegging until you can bring up any number in the list from 0-13 without needing to refer to the original source or other nodes on the list.
- 3) Repeat this exercise once a day for two weeks, after the first week spend some time trying to reform past dates or lists without referring to the source.

Information Compression



Information Compression

Information compression is another common method for storing large quantities of arbitrary information. The goal with information compression is to reduce the size of information so that it can be associated together in a logical way. I'll explore three main ways of doing this: mnemonics, picture linking and notes reduction.

Mnemonics

Mnemonics are devices that store several ideas together by using a phrase or word that can organize the information. In first aid, one mnemonic for treating a bleeding injury is RED. Rest the injured area, Elevate the injured area and apply Direct pressure. Similar mnemonics have been used for centuries to store information.

Using a mnemonic is as simple as collecting several ideas you want to compress together and picking a word or phrase you could use to organize them. Acronyms such as RED, NASA or USA are useful for storing several words or ideas under a single word.



The best mnemonics have an easily identifiable word to store the ideas.

Picture Linking

Picture linking can be seen as a cross between the link method and diagramming or visceralization. Here the goal is to link several ideas together by representing them in a single image, theme or concept. In my management class, I linked together the five forces model by creating a little doodle of the “five horsemen of the business apocalypse”. Each visually connected the ideas with the different parts of the original theory.

Another way to use picture linking is to create one picture that combines several ideas together. Once you create a symbol to represent each idea, you put them all in the same vivid mental image.

I find picture linking works best if it is done on paper, creating small visual drawings to connect several ideas together at once. If done quickly, it can be faster than the link or peg method.



Notes Compression



Notes compression is a useful way to get a grasp of a lot of material. By itself, it isn't an entirely useful technique. But it can serve as the starting point for using other information compression techniques, linking, pegging or holistic learning methods. The idea behind notes compression is taking a huge quantity of

information you need to learn, and reducing it to just a few pages of notes. Note compression allows for two things:

1) **It allows you to organize large amounts of information.** Normally, it is difficult to take the hundreds of facts, concepts and ideas over the duration of a course and sort it fully. But if that information is represented on just two or three pages, it becomes easier to organize the ideas at the highest level.

2) **It allows for easier connections than would be previously possible.** Looking at the entire structure of your subject can allow you to draw connections and link information that would be difficult to do from a zoomed-in perspective.

Steps to Compress Your Notes

1) **Give yourself several blank sheets of white paper.** Take all the notes for the particular course or sub-unit you want to compress. This exercise can take an hour or two, so make sure you have the time.

2) **Starting with the smallest writing you can use, write down a major idea from your notes.** Use as few words as possible.

3) **Next to that idea, write a related idea, formula, concept or definition.** Write the idea in small lettering and use few words as a description to save space.

4) Continue this process of writing ideas in a loose format, **until you have written down, in reduced form, every major idea from your notes.** You should end up with a 1-3 sheets of paper with densely written information.

5) As an additional step, you may want to rewrite this page for better organization if your first attempt is difficult to sort through, or connections between the material are hard to see.

On the next page I have one half of a notes compression I did in a past course on statistics. This can help point out weaknesses and logically group ideas. Of course, you can compress notes from a fraction of a course or an entire topic, the scale is up to you.

Note Compression Example

(total ideas for one course grouped together and organized on one page)

STATS IN ONE PAGE

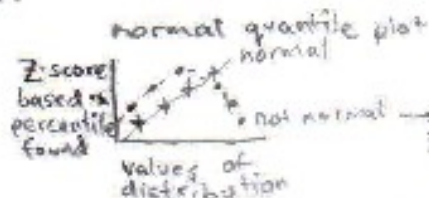
Mean isn't resistant, median is.

480

$$\text{Variance} = s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$Z_i = \frac{x_i - \bar{x}}{s}$$



VARS

y = response
x = explanatory

Correlation $r =$

$$\frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$D_r \{ r / r \in R, -1 \leq r \leq 1 \}$$

does not indicate causation

$|r| \approx 0$ (no linear correlation)
 $|r| \approx 1$ (strong correlation)

r isn't resistant

Least Squares Regression Line



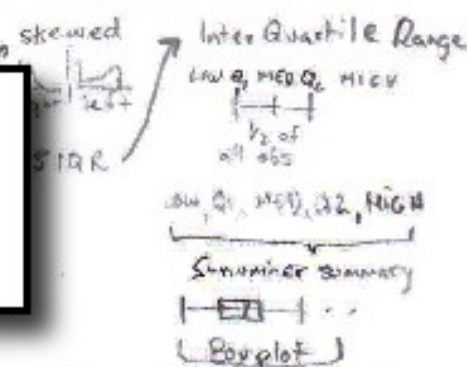
r^2 is d^2 of distribution explained by regression line



residual = $y - \hat{y}$
residual plots



Quantile = any percentage of a distribution



Approx Binom w/ Normal:

if $np \geq 10$ & $n(1-p) \geq 10$,

$$\frac{P}{n} \approx \frac{\hat{P}}{n}$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$$\mu = np$$

$$\sigma = \sqrt{np(1-p)}$$

$$\bar{x} \approx \mu$$

$$s_x \approx \frac{\sigma}{\sqrt{n}}$$

$$x \sim n(\mu, \sigma)$$

$$\bar{x} \sim n(\mu, \frac{\sigma}{\sqrt{n}})$$

Multi-stage



Sample

Experiment

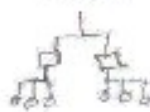
units/subjects

treatments

randomize

Block Design

units



Matched Pairs



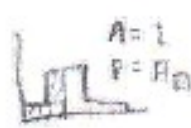
3 Principles: randomize, control, repetitions

Avoid Bias! & factors & levels

PRODUCING DATA

Observational Studies = Surveys

Specialized Random Survey



Confidence



Mind Challenge!

This mind challenge should get you used to using information compression methods.

- 1) Once per day, go through your textbook or notes and pick out a group of facts you would like to compress. Then create either a mnemonic or picture linking to group the ideas under one theme.
- 2) After doing this for two weeks, create a notes compression for the last two weeks of material, incorporating the various picture linking or mnemonics you used earlier.
- 3) Test yourself to see if you can remember the entire group of ideas after using one of these methods to compress several ideas together. Keep doing this process for four weeks (2 notes compressions) or until you feel comfortable with using these methods quickly and easily.

Extending Ideas

Just passing the test isn't enough. Having an A or A+ in your subjects isn't important if you can't use any of the information you've learned. The final stage in holistic learning is application. This is where you've not only understood and explored ideas, but you have tested, debugged and applied this knowledge.

Fully taking an idea from the acquiring to application stage isn't always necessary to get a good grade. But the further you can take an idea along the learning process, the more useful and memorable it becomes. None of these extension ideas are original, but they are easily forgotten. By extending ideas beyond what you need to get a grade, hopefully you can learn something worthwhile.

Practical Usage



Practical Usage

The best way to extend ideas you learn in courses is to apply the material in your practical life. This is rarely a problem with self-education or career-focused education. However, if you need to take many abstract courses to work towards a degree, some of the information may seem unnecessary or at least not applicable for years into the future. By making the ideas practical, they are more likely to stick.

Practical usage isn't a step-by-step technique. It is a creative process of looking for unusual ways you can apply the ideas you are ingesting. Although you might not be sure how you can use your history or calculus class in daily life, I'm sure you could find some uses if you spent ten minutes doing a brainstorm.

Instead of giving steps, I'd like to share some ways I've been able to use different ideas in courses I've studied. Some of these are highly practical solutions, others are strictly internal solutions by applying abstract course material to my own life.

1) **Statistics** - I used statistics courses to help decide the name of this book and the price to sell it at. Using Google AdWords, I ran an experiment testing a variety of names and prices. Statistics helped me pinpoint the significance of the sample size I found to see which should make it to the final product.

2) **Computer Science** - Aside from the obvious uses of programming, I've found computer science as a useful way of looking at problems. Debugging, algorithms, writing out procedures and organization are all useful skills that can be applied elsewhere.

3) **Accounting** - Taking accounting courses helped me sort out my personal finances and income statements. Applying the basic accounting principles to my finances made them easier to read and understand.

4) **Economics** - Economics helped me reevaluate how I look at money in society. By viewing it as a medium for exchanging value, I changed much of my philosophy for how money, the economy and world issues function.

5) **History** - History can be a tool for understanding the present. Learning about ancient Asian history helped me when trying to look at issues facing modern China, India and Japan. History provides me with a context for understanding world issues.

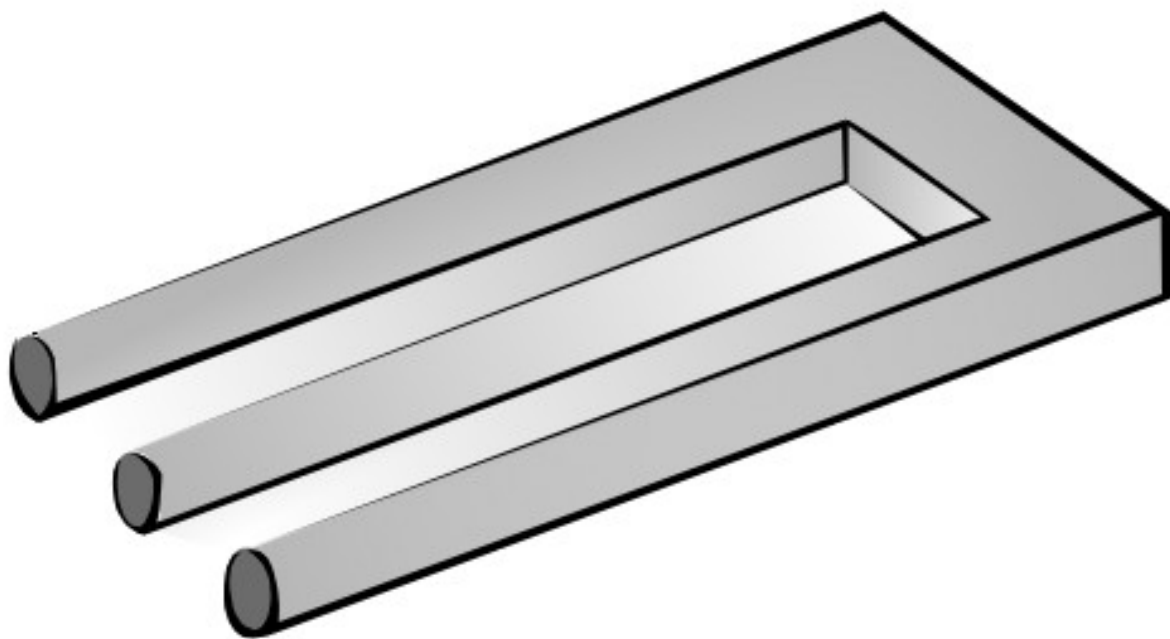
These are just a few examples, but I strive to make use of anything I can learn in subjects. If a course has absolutely no merit in daily life, what is the point of taking it?



Mind Challenge!

Once a week for the next two months, pick one of your major courses. Spend at least 5-10 minutes creating a list of ways you can use this information in your daily life. Make sure the list has at least 20 ideas and don't censor yourself in letting them flow. After you're done this weekly exercise, pick one of those ideas and implement it.

Model Debugging



Model Debugging

Model debugging is a fancy name for a simple concept: practice. Practicing questions and testing yourself is the only way you can get rid of errors in the holistic learning process. You can't debug a computer program easily without actually running it and looking for errors. Similarly, you can't debug your brain without testing it against questions and homework from your studies.

If you do the holistic learning process well, spending hours each day debugging shouldn't be necessary. Just like a skilled programmer can write more code with fewer errors, having the right skills in place from the beginning should reduce the time you spend practicing. Still, perfection is impossible, so taking the time to practice and debug your understanding is always a good idea.

Since you probably already know how to set up and do practice questions and homework, I won't repeat the obvious. But here are a few tips for improving the model-debugging process:

1) Separate Typos From Concept Errors

I'm sure most computer programmers are used to occasionally failing to compile

because they mistyped a key word. Worse is spending twenty minutes tracking down a bug because you used the wrong variable in an innocent mistake. But if you don't understand how a fundamental concept or algorithm should work, this is a far more serious problem.

The same applies to your own model debugging. When you get questions wrong, you need to separate simple errors that can easily be fixed, from those that represent a failure to understand key ideas. In the case of simple errors, just repeat a new question and learn from it. In the case of concept errors, go back to your models, metaphors and visceralizations to see if you made mistakes in the reasoning process.

2) The Shotgun Approach

Spending hours repeating the same type of questions is no different from rote memorization. I prefer to spend my time debugging on a shotgun approach, which tackles a few questions of each type, but doesn't become repetitive. I feel if you must spend hours repeating a question, then do it. But a better solution is to use the holistic learning methods earlier, so remembering how to solve certain problems become easier.

3) Spread Practice Times Out

Split your practice sessions into daily intervals, rather than pre-test crash sessions. A quick refresh each day will be more likely to stick than a huge download into your brain the night before.



Quick Tip!

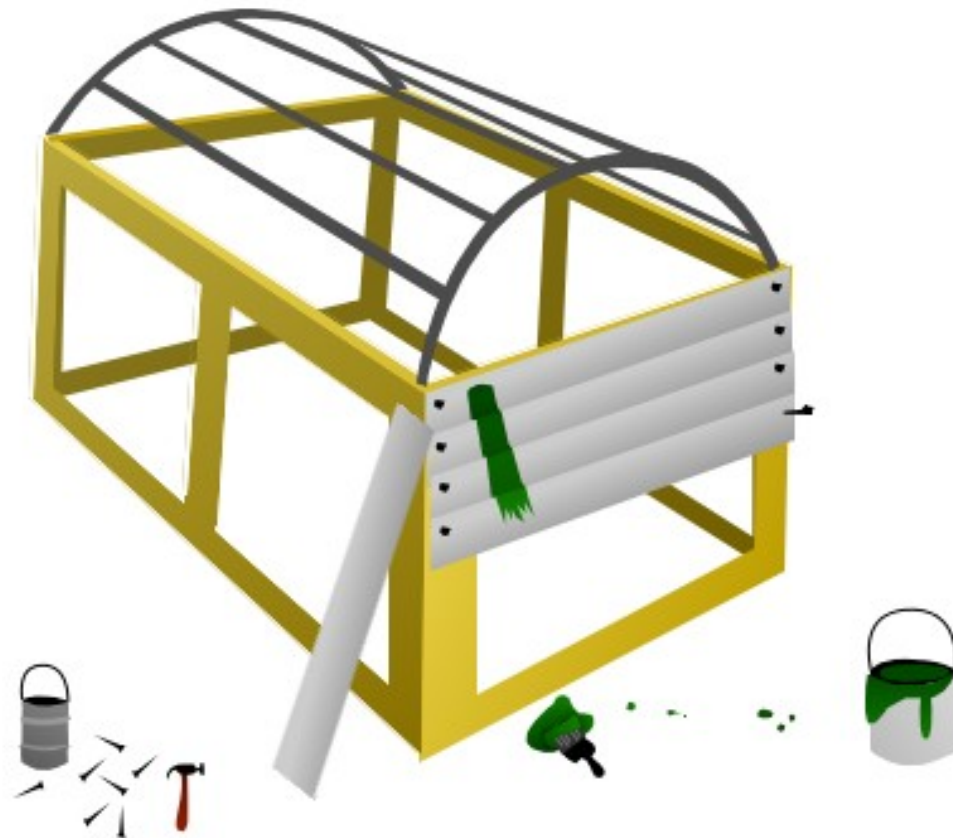
Practicing questions repeatedly is a slow way to learn. Model debugging is necessary, but if you need to do the same types of questions repeatedly, chances are you need more work on the exploring or understanding phases. If you feel you don't quite understand a topic, try going back and exploring the ideas.



Mind Challenge!

If you are already doing regular practice sessions for questions in your classes, continue with what you are doing. Try to implement some of these ideas for improving how you practice, but otherwise keep testing yourself. If you aren't doing regular practice sessions, or you are doing crash sessions before a test, try spreading your learning out over the entire term. Spend a few minutes each day instead of hours before the test.

Project-Based Learning



Project Based Learning

Project-based learning is my favorite technique for self-education. While going to this level of depth is probably unnecessary for most formal schooling, it can be useful when you don't have an instructor or a final exam.

Project-based learning simply means constructing a 1-3 month project that will force you to learn. In order to complete the project, you must know more than you do. Taking on a project will drive you to learn as you go, pushing any new ideas right through every step of the holistic learning process.

Taking on a 1-3 month project can be a little daunting, but if you are interested in really knowing a subject, I believe it is the best way to learn quickly. As with the last two methods for extending ideas, this one doesn't have a step-by-step formula. But here are some examples you might want to follow for learning different subjects:

1) **Computer Programming** - Design a project that will take you 1-3 months of work to complete. This could be a small application, a website or game.

2) **History** - Write a short e-book about the historical subject you want to study. Write one that will get you to do research as well as create your own thesis.

3) **Financial Accounting** - Pick several companies and do a complete analysis of their financial statements before deciding who you would invest money in.

Before you start designing a project to encourage your self-education, here are some tips I've found useful when running my own projects:

1) **Keep it Small.** A project should push your abilities forward, but massive, year-long adventures can start to drag after several months. If learning is your goal, keep the projects within a few months.

2) **Write it Down.** Commit to your project on paper, otherwise it won't last. I don't recommend committing to projects if you seriously doubt your motivation is strong enough to last. However, written documents can help you stay focused if you feel frustrated or bored by the next immediate activity.

3) **Create an Objective Outcome.** Just working on a project isn't always enough. I've found my learning efforts work best if there is a clear goal at the end of the project. If the project serves a function beyond just learning, you can boost your motivation towards continuing it.



Mind Challenge!

Design a project that you expect to take one month to accomplish. Examples could include:

- E-book
- Invention
- Design
- Program
- Website
- Blog

The project should be simple, but have a sharp learning curve. Write out your deadline for completion and schedule a bit of time each day to work on this project. After you've finished the project do a review of the project-based learning method and decide what other projects you might like to tackle.

Review of Holistic Learning Techniques

Theory and technique are equal parts to the holistic learning strategy. With just the theory, it is difficult to put holistic learning into practice. With just the techniques, any methods you use will be haphazard and random at best.

These techniques just represent a small minority of the total ways you can learn. I encourage you to find other methods and incorporate them into your approach. Use what works best for you and discard everything else until you find a pattern that works.

Practice is essential for making use of these holistic learning methods. That is why I have created a Mind Challenge for each of the different methods. The purpose of these challenges is to focus you on practicing one technique long enough for it to become a habit. Just reading about these ideas isn't enough if you can't use them automatically in your studies.

Completing all of the challenges would take close to two years and might not be feasible for the immediate future. I suggest that you pick only one challenge to work on in the next month or two. Base your decision on which method you would turn into a habit. If your obstacles are in understanding concepts, you might want to practice metaphor. If you can't handle the volume of dates and definitions, linking or pegging might be a better method to master.

The holistic learning theory should help you pinpoint where you might want to improve. From that point, you can determine which techniques would improve on that weak point. By finally moving that technique from a tool, which requires effort to use, to a habit which happens automatically, you can improve the way you learn information.

Included with this e-book are several documents for practicing some of the techniques mentioned in the book. You can use these documents to help you get started with your practicing sessions.

◦ Part III ◦

Beyond Holistic Learning

The Productive Student



The Productive Student

Throughout this book I've emphasized "Learn More". Holistic learning expands your ability to handle difficult subjects, raising the upper limit for how well you can understand your classes. Learning holistically will help improve your grades and understanding, but it probably won't shave off huge amounts of studying time. The new methods can take a few weeks to master, so the time-saving benefits are long-term.

Now I'd like to focus on the other half of this book's title, "Study Less". Becoming a productive student or self-learner is the art of cutting down the amount of time you spend in the books. During school, I have time to run this business, write as much as 7000 words each week, exercise and run a Toastmasters club on top of full-time classes. Despite this schedule, I still have time to spend time with friends and enjoy weekends.

Here are the key ideas for becoming more productive as a student:

- | | |
|------------------------|------------------|
| 1) Energy Management | 4) Batch |
| 2) Don't "Study" | 5) Get Organized |
| 3) End Procrastination | |

Productivity Tip #1

Manage Your Energy

I've seen the victims of poor energy management. These are the students completely burnt out from excess projects, requiring injections of caffeine just to function. Balancing courses and life isn't easy, especially if you have a full schedule.

Your body has fuel cells, both physically and mentally. You can't run on empty forever. If you try to cheat your body into giving a few extra hours today, it can mean a dozen lost hours next week. If you've been struggling to cope with burnout, stress and exhaustion, you haven't been managing your energy.

Better energy management has two steps:

- 1) Increase your energy capacity.
- 2) Switch your schedule from a linear format to a circular format.

Increasing Your Energy

Energy isn't just a genetic gift. There are many things you can do to boost the amount of force you can give towards your work. Here are a few things to consider:

1) **Do you exercise 3-5 times per week?** If the answer is no, you're undercutting your potential energy levels. Unless you're unable to exercise for medical reasons, there is no excuse for not getting your 40 minutes a day. A small investment in exercise can have a huge impact on your ability to concentrate and work quickly.

2) **Do you get 7-8 hours of sleep each night?** Some people work best on 4-5 hours of sleep. But I'm willing to bet neither of us can do that. Pulling an all-nighter is a dangerous move when you need to keep your mind sharp.

3) **What do you eat?** Do you eat a diet high in meat, fat and sugar? Switch to a diet that emphasizes whole grains and unprocessed foods. This will keep your blood sugar levels even throughout the day, avoiding the highs and lows.

4) **How many glasses of water do you drink each day?** You are over two-thirds water. Dehydrating yourself is a quick way to zap your energy.

5) **What meals do you eat each day?** If your answer is lunch, dinner and sometimes breakfast, you're energy levels will suffer. Ideally, you should eat 4-5 smaller meals throughout the day to ensure a more constant supply of nutrients. Eating a bit right before bed can also help with morning grogginess by avoiding low-blood sugar when you wake up.

Circular Scheduling

A linear schedule has work spread evenly over the entire time period. A circular schedule focuses on doing a lot of work earlier with ample rest time afterwards. Keeping a circular schedule helps your body keep up a rhythm of high productivity instead of a death march. Here are few ways to make your schedule more circular:

1) **Set a day off each week.** I take one day off every week where I don't do any schoolwork. Compressing seven days of work into six might seem difficult at first, but taking a day off prevents burnout.

2) **Evenings off.** Compress all your work into the morning. Instead of taking breaks throughout the day, get your work done early so you can have a few hours off each night.

3) **90 minute timeboxes.** Set aside 90 minutes towards a project or studying. After the ninety minutes is complete, you stop working. Timeboxing can help you keep your studying time focused.



Quick Tip!

One of the best books on the subject of energy management is **[The Power of Full Engagement](#)**. Although I've briefly touched on the subject, this book can give far more depth. The book divides up energy into four categories and gives more advice for how to maximize your energy and work within cycles. A must-read for anyone serious about enhancing their productivity.

Productivity Tip #2

Don't "Study"

I never study. I'll read my textbooks. I'll review notes. I'll finish assignments and do practice questions. But I never "study". A big waste of your time is to spend time "studying" without specifically defining what you want to accomplish. At the end of the year, grades, not how many hours you "studied" are going to matter most.

Cal Newport, author of *How to Win at College*, claims he hates the word studying:

"It's ambiguous, and for most students it's entangled in all sorts of emotional baggage. They feel guilty if they haven't suffered enough in the library. Student life, for them, becomes a constant struggle — always trying to "study" more, yet always falling a bit short. It's a lot like dieting. But with less involvement from Kirstie Alley."¹

1. Cal Newport: "Studying with Kirstie Alley, Decoding the Quarter System, and Coping with Early Classes", <http://calnewport.com/blog/?p=258>

Instead of studying, define the activities you need to perform in order to learn the material. For myself these are the activities I need to do in order to get an A or A+ in most classes:

- 1) Read the chapters covered
- 2) Complete assignments and take notes in class
- 3) Use some of the holistic learning methods to tackle trouble spots.
- 4) Optionally take a set of flow-based notes for the material covered before tests.

That's it. My list has been narrowed down by doing as much learning in-class as possible, so your list might have more items. The point isn't that I have a short list, but that I have made a list in the first place. Unless you itemize what **you need to do** in order to learn, you'll waste endless hours "studying" instead of optimizing your time to actually learn.

Productivity Tip #3

Nuke Procrastination

If an assignment is due next Wednesday are you the type of person who starts working on it:

- (A) As soon as it is assigned.
- (B) All in one session, some time between today and the due date.
- (C) Tuesday night.
- (D) Wednesday morning, *ten minutes before class!*

Unfortunately, most students I know would have to pick either C or D. Which is the best answer? It may surprise you that I don't believe answering A is ideal either. As I'll explain, beating procrastination doesn't mean just completing work earlier.

The correct answer (in most cases) would actually be B. Completing an assignment in one go saves time by batching. In addition, if you complete the assignment using the Weekly/Daily Goals method, you can save yourself the stress and guilt of deciding when to do assignments.

Weekly/Daily Goals System

The W/D Goals method is one of the best ways I know of to combat procrastination. The idea is simple:

1) At the end of each week, compile a list of all the assignments, homework, reading and studying activities you want to do in the following week.

Unless something unexpected arises during the week, you are obligated to finish this list—but no more than this list. This splits off the endless amount of assignments and work you could be doing into a manageable chunk of one week. If you have a particularly busy week, you might hold-off on long-term assignments. If you have a lighter week, you might get ahead on reading for tests far before the date.

2) **At the end of each day, check your weekly to-do and create a daily goals list.**

The next step is to break down your weekly work over each day. That's six days of work (assuming you're taking a rest day). You're obligated to finish this list each day, but no more than this list.

What does the W/D Goals system accomplish?

1) It **saves you the stress** of deciding whether to work more or less on a day. You just check your list to see whether you're done.

2) It **keeps you from procrastinating** on big projects. By relying on your weekly and daily goals lists, instead of due dates, you are the one in control over your schedule.

3) It **helps you balance workloads**. By looking ahead at your schedule you can adapt your weekly and daily goals to smooth over your work. Instead of pulling all-nighters before a test, and wasting hours of your time three weeks before, you can split up the work.



[My Lists](#) This list: [Edit](#) | [Reorder](#) | [Share](#)

Here is a sample of the weekly/daily goals lists I maintain:

Weekly Goals: Feb 4 - Feb 10

- ☐ Weekly Blog Work
- ☐ PTB Article
- ☐ "Flex" Article
- ☐ Backup Website
- ☐ Read Thursday's ENT Case
- ☐ Read Next Tuesday's ENT Case
- ☐ Write Extra LMSL Chapters: "Not

I use [TadaList](#), which is an online to-do list program for storing my lists. Checking my Daily Goals regularly and my Weekly Goals each night, helps me stay on track.



[My Lists](#) This list: [Edit](#) | [Reorder](#) | [Share](#)

Daily Goals - Feb 5, 2008

- ☐ Classes
- *** ☐ Gym
- ☐ TM - Speech
- ☐ Read Accounting Ch 5
- ☐ Read Accounting Ch 6
- ☐ Read Thursday's ENT Case
- ☐ Comp Lab @ 11:30

Productivity Tip #4

Batching

Batching involves taking a group of similar, small sections of work and doing them at one time. Batching helps cut down studying time because you can focus completely instead of switching between tasks.

Between both schoolwork and writing, I use batching in many places. I might do all my required reading for each week at one time, or I might write 3-4 articles in a row. Here are some tips to add batching to your productivity system:

1) **Size matters.** Batching works best when collecting small sections of work together. You can't "batch" your entire course for the day before a final exam. Batching will have a negative impact if you take it beyond 3-5 hour sessions of work.

2) **One-Off Assignments.** If an assignment takes me less than 8 hours of work, I usually try to accomplish it in one sitting. Splitting up a 3 hour essay over 15 sessions can add hours of wasted time as you slowly build up momentum in your writing.

3) **Build Your Concentration Threshold.** Your concentration threshold is the amount of time you can focus on a task before your productivity takes a nosedive. By taking on larger batches of tasks, you can slowly expand your threshold. Having a high threshold is helpful because it means you can batch large sections of work.



Quick Tip!

Want to take batching further? Here are some ways you can put it into practice: [20 Tips for Batching to Save Time and Cut Stress.](#)

Productivity Tip #5

Be Organized

Being organized will not get you A's. I've known A+ students who have horrible organization skills and C students with flawless systems. Being organized can help with your productivity and learning, but don't assume that all your problems have disorganization at their root.

If you haven't already, here are the key steps necessary for staying organized:

- 1) **Everything has a home.** Tasks, assignments and due dates all get stored in a proper location. Mess is the result of the homeless. Devise a place to put everything, and it will be a lot easier to stay ordered.

2) **Carry a notepad with you at all times.** Keep something to write on at all times of the day. This is invaluable for writing down tasks, due dates or ideas.

3) **Maintain a calendar and a to-do list.** Use to-do lists to store tasks and projects. Use your calendar to store events and due dates.

Even though being organized isn't a cure-all solution, it can definitely help. Organization skills can make it far easier to be productive, while eliminating stress.



Quick Tip!

If you're struggling with poor organizing habits, I recommend reading [***Zen To Done***](#), by Leo Babauta. This book focuses on the hardest part of getting organized—turning those skills into habits. Babauta creates a simple framework for staying organized and getting work done.